

Home-Foreign Elasticity Functions and the Gains from Trade

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- Answer depends on the domestic expenditure share and a trade elasticity that is constant
- Same trade elasticity governs substitution btw home-foreign and across foreign sources
- Elasticity estimated from expenditure reallocation across foreign sources

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Theoretical Contributions:

- 1 GFT depend on a one-dimensional function of home-foreign elasticities as domestic share $\uparrow$
- 2 Identification under weak separability + common sector-specific function across countries

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Empirical Contributions:

- Directly estimate sector-level elasticity functions from changes in home expenditure relative to aggregate foreign expenditure
- Result over 1995-2007: Elasticities generally rise with domestic expenditure shares
 - ▶ e.g., fitted elasticity rises from 3.33 at sector-specific 25th percentiles to 7.01 at 75th percentiles

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Quantitative Contributions:

- Combine estimates + assumptions on sectoral aggregation and IO to measure GFT
- Mean GFT are 32 log pts; compare to 54 if each elasticity is fixed at initial level
- New Mechanism: A lower initial domestic expenditure tends to $\Rightarrow$ lower effective elasticity

Literature

- GFT sufficient statistics: ACR (2012), CR (2018), and Ansari et al. (2026)...
 - Directly estimate welfare-relevant elasticity function
 - Provide generalized quantitative Armington, Krugman, Melitz gravity models
-
- Nonparametric trade counterfactuals: Adao, Costinot, and Donaldson (2017); Adao, Arkolakis, and Ganapati (2026)
 - Identify the low-dimensional home-foreign factor demand system using tariff variation
-
- International macroeconomics: Backus et al. (1994), Heathcote and Perri (2002), Corsetti et al. (2008), Feenstra et al. (2018)...
 - Relative to Feenstra et al. (2018), allow the home-foreign elasticity to vary with the domestic expenditure share and estimate its function across countries and sectors

Gains from Trade

Environment

General price paths

The proportional foreign-price ray

Environment

Armington production

- Many countries $i \in \mathcal{N}$ endowed with L_i units of labor, linear production $y_i = a_i L_i$
- Each country produces one differentiated variety; markets are competitive and trade is balanced
- Let w_i denote the wage and $Y_i = w_i L_i$ nominal income
- Unit production cost $c_i = w_i / a_i$, iceberg trade cost $\mathcal{D}_{ki} \geq 1$, so $p_{ki} = c_i \mathcal{D}_{ki}$

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+ otherwise general homothetic preferences

- Country n has prefs over $x_n = (x_{ni})_{i \in \mathcal{N}}$ represented by homothetic utility function u_n
- Unit expenditure function $e_n(\mathbf{p}_n)$ so real income is $W_n \equiv Y_n / e_n(\mathbf{p}_n)$

A General Price Path Toward Autarky

- Let Γ be a path in consumer-price space, indexed by $t \geq 1$, beginning at the trade equilibrium
- Normalize every price by the home price

$$\pi_{ni}(t) \equiv \frac{p_{ni}(t)}{p_{nn}(t)}, \quad \pi_{nn}(t) = 1, \quad H_n(t) \equiv e_n(\pi_n(t))$$

- Let $\mu_{ni}(t) \equiv X_{ni}(t)/X_{nF}(t)$ denote origin i 's share of foreign expenditure
- Define the cumulative foreign relative-price change

$$z_n(t) \equiv \int_1^t \sum_{i \neq n} \mu_{ni}(u) \frac{d \ln \pi_{ni}(u)}{du} du$$

- Each foreign relative-price change is weighted by its current share of foreign expenditure

The Welfare Effect of a Small Move Along the Path

- Shephard's lemma gives

$$d \ln H_n(t) = [1 - \lambda_{nn}(t)] dz_n(t)$$

- At equilibrium endpoints, $Y_n/p_{nn} = a_n L_n$, so

$$W_n(t) = \frac{a_n L_n}{H_n(t)}$$

- Therefore

$$d \ln W_n(t) = - [1 - \lambda_{nn}(t)] dz_n(t)$$

- Welfare responds to the foreign expenditure share times the expenditure-share-weighted foreign relative-price change

The Home-Foreign Elasticity Along the Path

Assume $z'_n(t) > 0$ and define the path-specific home-foreign elasticity

$$\varepsilon_n^\Gamma(t) \equiv \frac{d \ln[X_{nn}(t)/X_{nF}(t)]/dt}{z'_n(t)}$$

This is both the Hicksian and Marshallian elasticity of the home-to-foreign expenditure ratio along Γ

$$d \ln \frac{\lambda_{nn}(t)}{1 - \lambda_{nn}(t)} = \varepsilon_n^\Gamma(t) dz_n(t)$$

If $\varepsilon_n^\Gamma(t) > 0$, then $\lambda_{nn}(t)$ rises strictly and can index the path:

$$dz_n(t) = \frac{1}{1 - \lambda_{nn}(t)} \frac{d \ln \lambda_{nn}(t)}{\varepsilon_n^\Gamma(\lambda_{nn}(t))}$$

Gains from Trade Along a Price Path

For a path Γ from the trade equilibrium toward autarky,

$$\ln \frac{W_n^T}{W_n^A} = \int_1^\infty [1 - \lambda_{nn}(t)] z'_n(t) dt$$

If $z'_n(t) > 0$ and $\varepsilon_n^\Gamma(t) > 0$ for every finite t ,

$$\ln \frac{W_n^T}{W_n^A} = \int_{\lambda_{nn}^T}^1 \frac{d \ln \lambda}{\varepsilon_n^\Gamma(\lambda)}$$

- Assume home good alone can deliver one util $\Rightarrow$ finite & $\lambda_{nn}(t) \rightarrow 1$
- Every admissible path gives same GFT b/c endpoints are the same
- Path-specific elasticity functions can differ even though integrals coincide

The Proportional Foreign-Price Ray

- Apply a common multiplier $m \geq 1$ to every foreign price relative to the home price

$$\pi_{ni}(m) = m \frac{p_{ni}(1)}{p_{nn}(1)} \quad \text{for every } i \neq n$$

- Every foreign relative price changes by $d \ln m$, so

$$dz_n(m) = \sum_{i \neq n} \mu_{ni}(m) d \ln m = d \ln m$$

- Hence $z_n(m) = \ln m$ and

$$\varepsilon_n(m) \equiv \frac{d \ln [X_{nn}(m) / X_{nF}(m)]}{d \ln m}$$

- **Foreign expenditure shares drop out:** measuring GFT along this ray does not require recovering substitution across foreign sources
- The data contain non-proportional price changes, motivating weak separability

Weak Separability and Identification

Environment + GFT Result
Nonparametric Identification
Approximating the Elasticity Function

Weak Separability

- **Restriction:** Weak separability between home and foreign goods within each sector
- Provides a foreign aggregate with exact price index P_{nFs}
- The home-foreign expenditure ratio depends on source-specific foreign prices only through P_{nFs}/p_{nns} , not through the path
- $\Rightarrow$ **same within-sector home-foreign elasticity function applies across paths**

Details

Many sectors

- **Within sectors:** Weak separability btw home and foreign goods
- **Across sectors:** Weak separability, with a general homothetic aggregator combining the sectoral price indexes
- **Common sourcing within destination-sector cells:** All users buy the same sectoral composite, so total absorption has the same origin shares
- Define the within-sector home-foreign trade elasticity

$$\varepsilon_{ns}(\lambda_{nns}) \equiv \frac{\partial \ln(X_{nns}/X_{nFs})}{\partial \ln(P_{nFs}/p_{nns})}$$

- The foreign-index change is the expenditure-share-weighted foreign relative-price change

$$d \ln \frac{P_{nFs}}{p_{nns}} = \sum_{i \neq n} \mu_{nis} d \ln \frac{p_{nis}}{p_{nns}}$$

Gains from Trade with Many Sectors

- Apply the one-sector argument within each sector to obtain the sectoral cost of autarky

$$I_{ns} \equiv \int_{\lambda_{ns}^T}^1 \frac{d \ln \lambda}{\varepsilon_{ns}(\lambda)}$$

- I_{ns} is the change in the log sectoral unit expenditure function relative to the price of country n 's good in sector s
- If nominal income and all domestic sectoral prices move proportionally, aggregate GFT are

$$\ln \frac{W_n^T}{W_n^A} = \ln \frac{\mathcal{E}_n \left((e_{ns}^T \exp(I_{ns}))_{s \in \mathcal{S}} \right)}{\mathcal{E}_n \left((e_{ns}^T)_{s \in \mathcal{S}} \right)}$$

- The cross-sector aggregator $\mathcal{E}_n$ determines how the sectoral cost changes combine into aggregate welfare

Estimating Equation + Restriction

- **Cross-country restriction:** $\varepsilon_{ns}(\lambda) = \varepsilon_s(\lambda)$ within sector s

Estimating Equation + Restriction

- **Cross-country restriction:** $\varepsilon_{ns}(\lambda) = \varepsilon_s(\lambda)$ within sector s
- Taking a first-order approx around $\lambda_{nns,0}$ yields

$$\Delta \ln \frac{X_{nns}}{X_{nFs}} = \varepsilon_s(\lambda_{nns,0}) (Q_{ns}^{\mathcal{T}} + Q_{ns}^P) + u_{ns}$$

$$Q_{ns}^{\mathcal{T}} \equiv \sum_{i \in \mathcal{O}_{ns}} \mu_{nis,0} \Delta \ln(1 + \mathcal{T}_{nis}), \quad Q_{ns}^P \equiv \sum_{i \in \mathcal{O}_{ns}} \mu_{nis,0} (\Delta \ln p_{iis} - \Delta \ln p_{nns})$$

- $Q_{ns}^{\mathcal{T}}$: tariff component of the foreign-relative-price change
- Q_{ns}^P : initial-share-weighted foreign output-price change relative to home output price
- u_{ns} : demand shifts, unmeasured trade-cost changes, and approximation error

Restricted Cubic Spline Approximation

$$\varepsilon_s(\lambda) = \theta_{s0} + \theta_{g(s)1}\lambda + \theta_{g(s)2}R_{g(s)}(\lambda)$$

- **Three knots:** 10th, 50th, and 90th percentiles of the initial domestic share within group g
- Cubic between knots; linear below the first knot and above the last knot
- Sector-specific intercepts; share-dependent coefficients pooled within group $g(s)$
- Nested cases:
 - ▶ $\theta_{g2} = 0$: linear elasticity function
 - ▶ $\theta_{g1} = \theta_{g2} = 0$: constant-elasticity CES benchmark

Estimating Home-Foreign Elasticities

Parametric Specification and Estimating Equation

Elasticity function:

$$\varepsilon_s(\lambda) = \theta_{s0} + \theta_{g(s)1}\lambda + \theta_{g(s)2}R_{g(s)}(\lambda)$$

Estimating equation:

$$\begin{aligned}\Delta \ln \frac{X_{nns}}{X_{nFs}} &= [\theta_{s0} + \theta_{g1}\lambda_{nns,0} + \theta_{g2}R_g(\lambda_{nns,0})] Q_{ns}^T \\ &\quad + [\gamma_{s0} + \gamma_{g1}\lambda_{nns,0} + \gamma_{g2}R_g(\lambda_{nns,0})] Q_{ns}^P + \mathbf{A}'_{ns}\boldsymbol{\kappa} + u_{ns}\end{aligned}$$

- One time change removes time-invariant destination-sector demand heterogeneity
- Separate coefficients on Q_{ns}^T and Q_{ns}^P allow measured prices to be endogenous or imperfectly scaled

Variation Used in Estimation

$$Z_{ns}^{\mathcal{T}} = \sum_{i \in \mathcal{O}_{ns}} \mu_{nis,0} Z_{nis}^{\mathcal{T}}$$

- $Z_{nis}^{\mathcal{T}}$: tariff change implied by predetermined trade agreements; zero for other pairs
- Instrument each tariff regressor by replacing $Q_{ns}^{\mathcal{T}}$ with $Z_{ns}^{\mathcal{T}}$
 - ▶ Sector interactions identify elasticity intercepts
 - ▶ $\lambda_{nns,0} Z_{ns}^{\mathcal{T}}$ and $R_g(\lambda_{nns,0}) Z_{ns}^{\mathcal{T}}$ identify how elasticities vary with the domestic share
- Controls: flexible functions of Q_{ns}^P and additive functions of the initial domestic share
- Exactly identified, unrestricted linear 2SLS; SEs clustered by destination

[Moments](#) [Details](#)

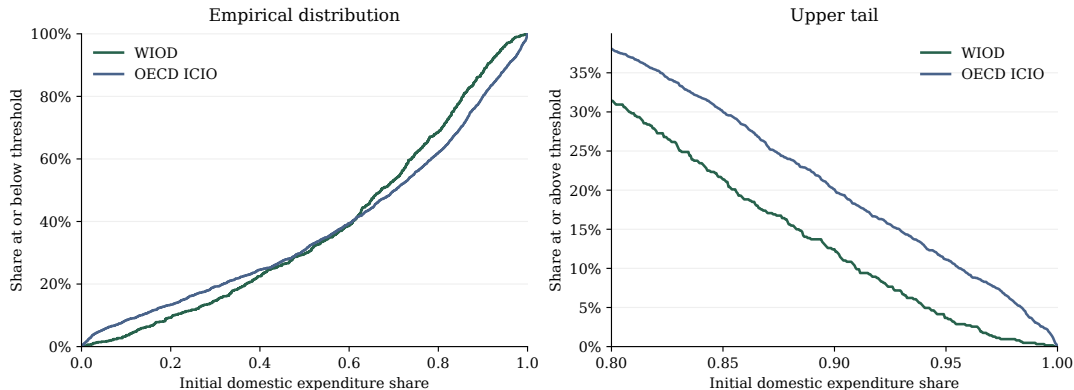
Data and Measurement

Domestic expenditure: Less standard or well measured than international trade flows → use two distinct sources

	WIOD	OECD ICIO
Role	Baseline	Replication
Time change	1995–2007	1995–2007
Destinations	40	80
Goods-producing sectors	16	26
Destination–sector observations	627	2,030

- Both measure total absorption, including intermediate and final uses
- Bilateral tariffs: Teti's Global Tariff Database
- Output prices: companion industry price series in each input-output source

Extrapolation: The Distribution of $\lambda_{nns,0}$



- **Observed support:** Both samples contain substantial variation, including domestic expenditure shares near one
- **Extrapolation:** Quantification evaluates each fitted function from observed share to one

Estimation Results: WIOD Baseline

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F
(1)	All goods	12.02 (4.29)	3.33 (1.22)	7.01 (1.76)	83.3%	4,066.58
(2)	Manufacturing	13.85 (5.70)	3.71 (1.33)	7.86 (2.13)	85.0%	2,162.62
	Non-manuf. goods	22.31 (13.02)	-1.30 (4.76)	5.32 (4.28)	72.5%	192.54
(3)	Consumer manuf.	15.65 (6.64)	1.11 (1.85)	5.32 (2.59)	68.0%	1,144.18
	Producer manuf.	24.03 (8.13)	4.76 (1.99)	12.00 (2.69)	87.8%	329.50
	Non-manuf. goods	22.31 (13.02)	-1.30 (4.76)	5.32 (4.28)	72.5%	192.54

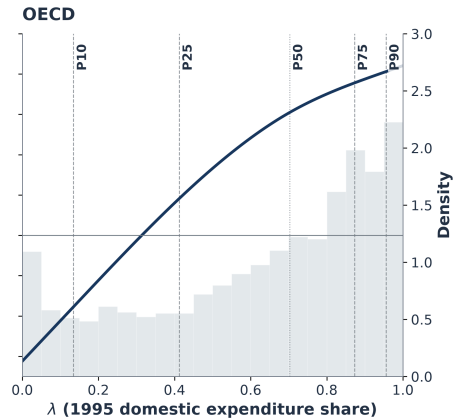
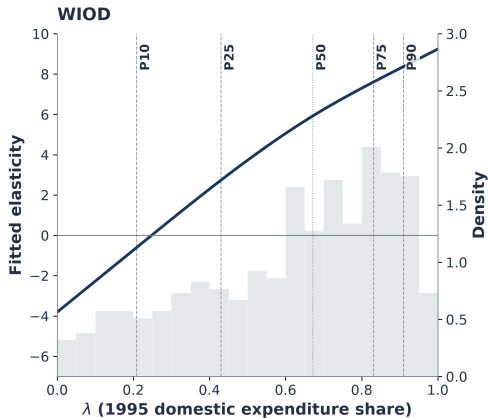
The all-goods average fitted elasticity rises from 3.33 at sector-specific 25th percentiles to 7.01 at 75th percentiles.

Replication: OECD ICIO

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F
(1)	All goods	11.72 (5.81)	2.14 (1.93)	6.89 (1.75)	72.9%	457.43
(2)	Manufacturing	9.06 (5.70)	1.07 (2.19)	5.20 (1.70)	68.2%	209.96
	Non-manuf. goods	43.55 (18.17)	3.40 (2.33)	15.43 (3.16)	80.0%	310.90
(3)	Consumer manuf.	13.45 (6.88)	0.43 (2.44)	6.26 (2.04)	74.5%	80.45
	Producer manuf.	3.66 (10.29)	2.04 (3.93)	3.93 (2.23)	61.6%	81.95
	Non-manuf. goods	43.55 (18.17)	3.40 (2.33)	15.43 (3.16)	80.0%	310.90

The OECD replication also implies positive average slopes, with less precise manufacturing estimates.

Estimated Elasticity Functions



- In both data sets, the fitted all-goods elasticity rises throughout the domestic-share distribution and flattens at higher shares

Estimator Validation and Empirical Sensitivity

Monte Carlo I: Estimator does not generate positive slopes when the true elasticity functions are flat

Monte Carlo II: Estimator correctly identifies positive slopes when true functions are linear

Details

1. Samples + controls

- Country characteristics
- Sector-specific additive controls
- At least 50% tariff-price coverage

Table
Table
Table

2. Tariff construction

- Trade-weighted applied tariffs
- Agreement-implied changes remain the excluded instruments

Table

3. Functional form + influence

- Quadratic function
- Leave-one-country/sector-out

Table
Figure

4. Positivity + finite changes

- Positive elasticity floor
- Finite tariff response
- Both restrictions
- Lagged-share IV

Table
Table
Table
Current Lagged

Average slopes remain positive across every reported group and every country or sector omission.

Positivity restrictions flatten the estimated functions.

Correlated Measurement Error?

- $\lambda_{nns,0}$ and the initial home-foreign expenditure ratio use the same expenditure data
- Measurement error can therefore be mechanically correlated across the dependent variable and the slope regressor
- Use 1995 shares to instrument every share-based regressor in an estimating equation based on 1996–2007 changes

	Slope at medians	$\widehat{\varepsilon}_{25}$	$\widehat{\varepsilon}_{75}$
Current-share interactions	11.09	2.21	6.42
Lagged-share instruments	11.13	2.42	6.15

- **Similar magnitudes suggest correlated M.E. is not driving the estimates**

Quantification

Production and Final Uses

- **Sectors:** manufacturing + non-manufacturing goods sectors; remaining sectors form a non-traded composite
- **Production:** Every sector in country n has value-added share β_n and uses the same aggregate composite as its intermediate input

$$c_{ns} = a_{ns}^{-1} w_n^{\beta_n} P_n^{1-\beta_n}$$

- **Final uses:** Household consumption, government purchases, and investment are current final demands; capital stocks are fixed
- **Intermediate inputs:** The scalar multiplier $1/\beta_n$ amplifies gains from final expenditure; median $\beta_n = 0.47$ in WIOD and 0.48 in OECD ICIO

Details

Preferences

- **Sectoral cost of autarky:**

$$\hat{e}_{ns} = \exp \left(\int_{\lambda_{nns}^T}^1 \frac{d \ln \lambda}{\varepsilon_{ns}(\lambda)} \right)$$

- **Lower tier:** CES across sectors within manufacturing and non-manufacturing goods, with $\sigma_L = 2$
- **Upper tier:** CES across manufacturing, non-manufacturing goods, and non-goods, with $\sigma_U = 0.6$

$$\ln \frac{W_n^T}{W_n^A} = \frac{1}{\beta_n(1 - \sigma_U)} \ln \left[\alpha_{nM} \hat{E}_{nM}^{1-\sigma_U} + \alpha_{nG} \hat{E}_{nG}^{1-\sigma_U} + \alpha_{n0} \right]$$

Quantitative Results

WIOD gains from trade: $100 \ln(W_n^T / W_n^A)$

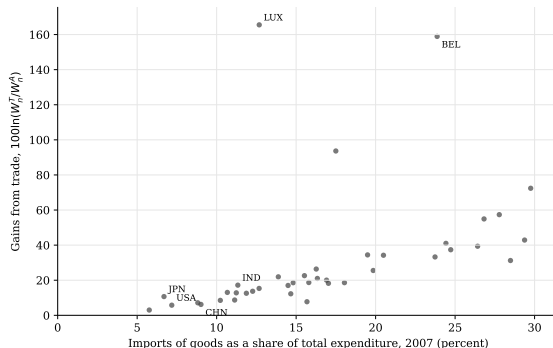
	p10	p25	Median	Mean	p75	p90
Specification (1)	5.75	10.21	15.50	21.00	30.66	39.04
Specification (2)	5.50	9.95	16.13	27.48	31.72	50.00
Specification (3)	7.68	12.75	19.37	32.00	35.17	58.87

- **Specification (3):** median gains are 19.37 log points and mean gains are 32.00
- Gains are substantial and heterogeneous across countries

OECD

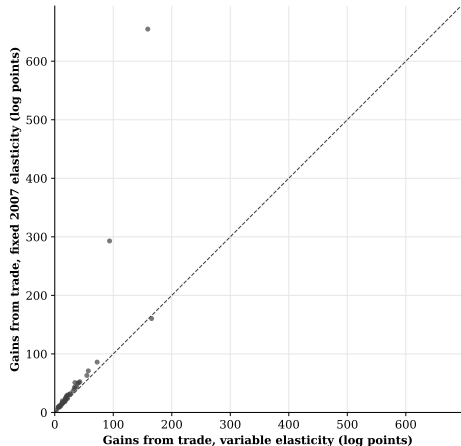
Within-sector mechanism: A lower initial domestic share means greater trade exposure, a lower starting elasticity, and a longer portion of the function traversed on the path to autarky. OECD

Gains from Trade and Initial Import Shares



- **Trade exposure:** Correlation is 0.42; countries w/ $\uparrow$ initial import shares generally gain more
- **Composition:** Countries with similar aggregate import shares can differ because sectoral shares, elasticity functions, and value-added shares differ
 - ▶ Luxembourg and Belgium: low domestic shares in non-manufacturing sectors w/ low elasticities

Variable and Fixed Home-Foreign Elasticities



- **Initial value:** Hold each country-sector elasticity fixed at its fitted 2007 value
- Median rises from 19.37 to 24.43 log points; mean rises from 32.00 to 54.42
- Fixed elasticities raise gains for 39 of 40 countries; Luxembourg is the exception

Sensitivity

σ_L	σ_U	p10	p25	Median	Mean	p75	p90
2	0.6	7.68	12.75	19.37	32.00	35.17	58.87
1	0.6	9.85	40.03	76.12	∞	521.98	∞
3	0.6	6.34	9.89	14.77	24.33	25.42	41.70
2	1	7.58	12.05	18.39	25.49	32.41	54.45

- Raising σ_L allows substitution away from sectors with the largest price-index increases and lowers measured gains
- With $\sigma_L = 1$, zero domestic expenditure makes gains infinite for nine countries
- Raising σ_U from 0.6 to one lowers the median more modestly, from 19.37 to 18.39

Conclusion

Conclusion

- **Negative:** Using an elasticity estimated from substitution across foreign sources either mismeasures GFT or requires strong assumptions equating distinct substitution margins
- **Positive:** The welfare-relevant home-foreign elasticity function is low dimensional and can be estimated directly
- **Evidence:** In WIOD and OECD ICIO, fitted elasticities generally rise with domestic expenditure shares, including under extensive sensitivity
- **Quantification:** In WIOD, mean GFT are 32.00 log points, compared with 54.42 when each elasticity is fixed at its initial value

Appendix

$$\Delta \ln \frac{X_{nns}}{X_{nFs}} = \varepsilon_s(\lambda_{nns,0}) (Q_{ns}^{\mathcal{T}} + Q_{ns}^P) + u_{ns}, \quad Q_{ns}^P = \sum_{i \in \mathcal{O}_{ns}} \mu_{nis,0} (\Delta \ln p_{iis} - \Delta \ln p_{nns})$$

For each $\lambda \in \text{supp}(\lambda_{nns,0} \mid s)$:

Condition on $\lambda_{nns,0} = \lambda$, Q_{ns}^P , predetermined shares and controls, and sector s . Assume

$$\underbrace{\text{Var}(Q_{ns}^{\mathcal{T}} \mid \cdot) > 0}_{\substack{\text{Relevance} \\ \text{(rank condition)}}} \quad \text{and} \quad \underbrace{\text{Cov}(Q_{ns}^{\mathcal{T}}, u_{ns} \mid \cdot) = 0}_{\text{Exclusion}}$$

These conditions identify

$$\varepsilon_s(\lambda) = \frac{\text{Cov}\left(\Delta \ln \frac{X_{nns}}{X_{nFs}}, Q_{ns}^{\mathcal{T}} \mid \cdot\right)}{\text{Var}(Q_{ns}^{\mathcal{T}} \mid \cdot)} \quad \text{for every } \lambda \in \text{supp}(\lambda_{nns,0} \mid s).$$

- Beyond the support of domestic expenditure shares in sector s , GFT relies on functional form to extrapolate the elasticity function

Write the estimating equation as

$$Y_{ns} = D'_{ns}\theta + W'_{ns}\delta + u_{ns},$$

where D_{ns} contains tariff regressors, W_{ns} contains measured-price and additive controls, and Z_{ns} contains the corresponding agreement instruments.

Residualize the excluded instruments with respect to the controls:

$$\tilde{Z}_{ns} = Z_{ns} - \mathbb{E}[Z_{ns} W'_{ns}] \mathbb{E}[W_{ns} W'_{ns}]^{-1} W_{ns}.$$

The population moments are

$$\mathbb{E}[W_{ns} u_{ns}] = 0, \quad \mathbb{E}[\tilde{Z}_{ns} u_{ns}] = 0.$$

Identification additionally requires

$$\text{rank } \mathbb{E}[\tilde{Z}_{ns} D'_{ns}] = \dim(\theta).$$

The elasticity coefficients are identified by agreement variation remaining after the measured-price and additive controls are partialled out.

- Construct the three spline knots within each group and evaluate $R_g(\lambda_{nns,0})$
- Create sector interactions with Q_{ns}^T and group interactions with $\lambda_{nns,0}Q_{ns}^T$ and $R_g(\lambda_{nns,0})Q_{ns}^T$
- Instruments replace Q_{ns}^T with Z_{ns}^T in every tariff interaction
- Controls include group share functions and flexible sector- and group-specific interactions with Q_{ns}^P
- Ordinary linear 2SLS; one excluded instrument for each tariff regressor
- Destination-clustered covariance; report the minimum conditional Sanderson–Windmeijer F statistic within each group

$$\hat{\theta} = (\tilde{D}'P_{\tilde{Z}}\tilde{D})^{-1}\tilde{D}'P_{\tilde{Z}}\tilde{y}$$

Appendix: Gains Without Monotonicity

The path-indexed welfare result does not require $z'_n(t) > 0$ or $\varepsilon_n^\Gamma(t) > 0$:

$$\ln \frac{W_n^1}{W_n^0} = - \int_{t_0}^{t_1} [1 - \lambda_{nn}(t)] z'_n(t) dt$$

This expression remains valid when the weighted foreign relative-price change or the domestic expenditure share reverses direction.

If the home good alone can deliver one unit of utility, then $\lambda_{nn}(t) \rightarrow 1$ and gains from trade are finite:

$$\ln \frac{W_n^T}{W_n^A} = \int_1^\infty [1 - \lambda_{nn}(t)] z'_n(t) dt.$$

If this condition fails, real income converges to zero and gains from trade are infinite.

Foreign relative-price growth along a path Γ from trade ($t = 1$) to autarky:

$$\alpha_{ni}(t) \equiv \frac{d \ln \pi_{ni}(t)}{dt} \quad (i \neq n) \quad \text{Proportional ray: } \alpha_{ni}(t) = \frac{d \ln m(t)}{dt}$$

With $\mu_{ni}(t) \equiv X_{ni}(t)/X_{nF}(t)$, the expenditure equation becomes

$$\frac{d \ln H_n(t)}{dt} = \sum_{i \neq n} \lambda_{ni}(t) \alpha_{ni}(t) = [1 - \lambda_{nn}(t)] \sum_{i \neq n} \mu_{ni}(t) \alpha_{ni}(t).$$

$$\text{Define } z'_n(t) \equiv \sum_{i \neq n} \mu_{ni}(t) \alpha_{ni}(t) \quad \text{Proportional ray: } z'_n(t) = \frac{d \ln m(t)}{dt}$$

The home-foreign elasticity along Γ gives

$$\varepsilon_n^\Gamma(t) \equiv \frac{d \ln [X_{nn}(t)/X_{nF}(t)]/dt}{z'_n(t)} \quad \Rightarrow \quad z'_n(t) = \frac{1}{[1 - \lambda_{nn}(t)] \varepsilon_n^\Gamma(t)} \frac{d \ln \lambda_{nn}(t)}{dt}.$$

Assume $z'_n(t) > 0$ and $\varepsilon_n^\Gamma(t) > 0$, so $\lambda_{nn}(t)$ can index the path. Combining gives

$$\frac{d \ln H_n(t)}{dt} = \frac{1}{\varepsilon_n^\Gamma(t)} \frac{d \ln \lambda_{nn}(t)}{dt} \quad \Rightarrow \quad \ln \frac{W_n^T}{W_n^A} = \int_{\lambda_{nn}^T}^1 \frac{d \ln \lambda}{\varepsilon_n^\Gamma(\lambda)}.$$

- Within sector s , weak separability yields a foreign aggregate with exact price index P_{nFs}
- Holding preferences fixed,

$$d \ln \frac{P_{nFs}}{p_{nns}} = \sum_{i \neq n} \mu_{nis} d \ln \frac{p_{nis}}{p_{nns}}$$

- The home-foreign expenditure ratio depends only on P_{nFs}/p_{nns} :

$$\frac{X_{nns}}{X_{nFs}} = r_{ns} \left(\frac{P_{nFs}}{p_{nns}} \right)$$

- Therefore

$$\varepsilon_{ns}(\lambda_{nns}) \equiv \frac{\partial \ln(X_{nns}/X_{nFs})}{\partial \ln(P_{nFs}/p_{nns})}$$

is the same function along proportional and non-proportional source-price paths

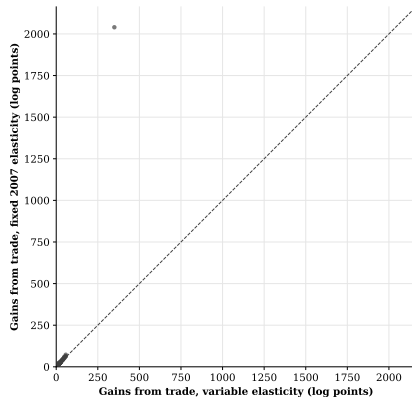
- This connects the responses identified from observed tariff variation to the function used in the gains-from-trade calculation

- Every production sector in country n has the same value-added share β_n
- Every sector uses the final-consumption aggregate as its intermediate-input composite
- A one-percent rise in the final-expenditure price index raises production costs by $1 - \beta_n$ percent
- This feedback amplifies the final-expenditure gains by $1/\beta_n$

$$\ln \frac{W_n^T}{W_n^A} = \frac{1}{\beta_n} \times \text{change in the final-expenditure price index}$$

	p10	p25	Median	Mean	p75	p90
Specification (1)	8.19	10.73	16.72	20.69	25.77	37.80
Specification (2)	14.62	17.98	25.12	33.31	35.22	54.95
Specification (3)	10.96	13.50	19.59	26.90	27.48	44.00

- Entries are $100 \ln(W_n^T / W_n^A)$ across 80 countries
- Under Specification (3), median GFT are 19.59 log points and mean GFT are 26.90



- Mean GFT are 26.90 log points under variable functions and 50.46 when each country-sector elasticity is held fixed at its initial value

- Simulate tariff shocks in a multi-sector extended Armington model aligned with the WIOD baseline
 - ▶ nested homothetic CDE-CES expenditure system
 - ▶ separate home-foreign vs. across foreign substitution
 - ▶ home-foreign elasticity function is exactly linear
- Calibrate four versions of the model to the same initial equilibrium
 - ▶ Constant elasticities [Table](#)
 - ▶ Common true slope of four [Table](#)
 - ▶ Manufacturing slope of three and non-manufacturing slope of six [Table](#)
 - ▶ Consumer, producer, and non-manufacturing slopes of three, five, and seven [Table](#)
- Apply the baseline spline estimator to 200 independently drawn tariff-shock matrices

Appendix: Monte Carlo, Constant Elasticities

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Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	0.000	0.002	[-0.004, 0.008]	67.5%
Manufacturing	0.000	0.001	[-0.004, 0.008]	60.5%
Consumer manuf.	0.000	0.003	[-0.006, 0.012]	66.0%
Producer manuf.	0.000	-0.001	[-0.012, 0.011]	45.5%
Non-manuf. goods	0.000	0.000	[-0.028, 0.031]	49.0%

Notes: Each row summarizes 200 simulated economies estimated with the unrestricted three-knot spline. Est. mean averages across draws the derivative evaluated at each sector's median initial share and then equally across sectors. True mean gives each sector equal weight. When true slopes differ within a row, this is a descriptive benchmark; the pooled coefficient need not estimate that average. The percentile interval describes the simulation distribution and is not a confidence interval. Share > 0 is the fraction of simulated average-slope estimates above zero. Elasticity and measured-price-control intercepts vary by sector. The additive-control intercept and all share-dependent coefficients vary by group. The model has 40 countries and 16 goods sectors: seven consumer-manufacturing, seven producer-manufacturing, and two non-manufacturing sectors. A draw's tariff shocks are held fixed across calibrations.

Appendix: Monte Carlo, Common Positive Slope

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Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	4.000	4.027	[4.015, 4.039]	100.0%
Manufacturing	4.000	4.024	[4.011, 4.038]	100.0%
Consumer manuf.	4.000	4.017	[4.002, 4.036]	100.0%
Producer manuf.	4.000	4.032	[4.008, 4.056]	100.0%
Non-manuf. goods	4.000	4.050	[3.989, 4.119]	100.0%

Notes: Each row summarizes 200 simulated economies estimated with the unrestricted three-knot spline. Est. mean averages across draws the derivative evaluated at each sector's median initial share and then equally across sectors. True mean gives each sector equal weight. When true slopes differ within a row, this is a descriptive benchmark; the pooled coefficient need not estimate that average. The percentile interval describes the simulation distribution and is not a confidence interval. Share > 0 is the fraction of simulated average-slope estimates above zero. Elasticity and measured-price-control intercepts vary by sector. The additive-control intercept and all share-dependent coefficients vary by group. The model has 40 countries and 16 goods sectors: seven consumer-manufacturing, seven producer-manufacturing, and two non-manufacturing sectors. A draw's tariff shocks are held fixed across calibrations.

Appendix: Monte Carlo, Two Group Slopes

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Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	3.375	3.230	[3.158, 3.317]	100.0%
Manufacturing	3.000	3.021	[3.011, 3.033]	100.0%
Consumer manuf.	3.000	3.017	[3.004, 3.033]	100.0%
Producer manuf.	3.000	3.026	[3.006, 3.047]	100.0%
Non-manuf. goods	6.000	6.070	[5.994, 6.157]	100.0%

Notes: Each row summarizes 200 simulated economies estimated with the unrestricted three-knot spline. Est. mean averages across draws the derivative evaluated at each sector's median initial share and then equally across sectors. True mean gives each sector equal weight. When true slopes differ within a row, this is a descriptive benchmark; the pooled coefficient need not estimate that average. The percentile interval describes the simulation distribution and is not a confidence interval. Share > 0 is the fraction of simulated average-slope estimates above zero. Elasticity and measured-price-control intercepts vary by sector. The additive-control intercept and all share-dependent coefficients vary by group. The model has 40 countries and 16 goods sectors: seven consumer-manufacturing, seven producer-manufacturing, and two non-manufacturing sectors. A draw's tariff shocks are held fixed across calibrations.

Appendix: Monte Carlo, Three Group Slopes

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Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	4.375	4.140	[3.984, 4.307]	100.0%
Manufacturing	4.000	3.916	[3.769, 4.081]	100.0%
Consumer manuf.	3.000	3.017	[3.004, 3.033]	100.0%
Producer manuf.	5.000	5.035	[5.008, 5.062]	100.0%
Non-manuf. goods	7.000	7.081	[6.993, 7.178]	100.0%

Notes: Each row summarizes 200 simulated economies estimated with the unrestricted three-knot spline. Est. mean averages across draws the derivative evaluated at each sector's median initial share and then equally across sectors. True mean gives each sector equal weight. When true slopes differ within a row, this is a descriptive benchmark; the pooled coefficient need not estimate that average. The percentile interval describes the simulation distribution and is not a confidence interval. Share > 0 is the fraction of simulated average-slope estimates above zero. Elasticity and measured-price-control intercepts vary by sector. The additive-control intercept and all share-dependent coefficients vary by group. The model has 40 countries and 16 goods sectors: seven consumer-manufacturing, seven producer-manufacturing, and two non-manufacturing sectors. A draw's tariff shocks are held fixed across calibrations.

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	13.07 (5.77)	2.79 (1.92)	6.60 (2.21)	77.5%	561.25	627
(2)	Manufacturing	14.04 (6.62)	3.37 (2.09)	7.53 (2.73)	80.4%	519.86	547
	Non-manuf. goods	29.27 (15.00)	-1.42 (5.30)	6.23 (4.73)	63.7%	519.86	80
(3)	Consumer manuf.	15.37 (7.23)	1.08 (2.66)	5.25 (3.42)	66.9%	717.91	269
	Producer manuf.	35.34 (11.31)	3.27 (2.99)	13.38 (3.37)	83.1%	717.91	278
	Non-manuf. goods	34.84 (15.37)	-2.53 (5.36)	6.42 (4.79)	58.8%	717.91	80

Notes: This table modifies Table 1 by allowing destination characteristics to enter the elasticity intercept and the coefficient on measured-price changes, with coefficients common across groups. The reported quartile elasticities also average the characteristic contribution across destinations within each sector. Min. SW F is calculated for the entire joint regression and therefore repeats across its group rows. Non-manufacturing estimates can differ across Specifications (2) and (3).

Spec.	Aggregate	Average slope	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	14.14 (6.53)	3.25 (1.14)	7.36 (2.13)	79.7%	874.49	627
(2)	Manufacturing	15.07 (7.58)	3.70 (1.37)	8.12 (2.59)	81.4%	695.00	547
	Non-manuf. goods	30.35 (15.17)	-1.01 (4.77)	8.87 (8.16)	72.5%	15.11	80
(3)	Consumer manuf.	17.59 (7.95)	0.84 (1.92)	5.85 (3.02)	66.2%	398.72	269
	Producer manuf.	28.60 (14.01)	4.35 (1.78)	12.61 (3.94)	85.3%	180.26	278
	Non-manuf. goods	30.35 (15.17)	-1.01 (4.77)	8.87 (8.16)	72.5%	15.11	80

Notes: This table modifies Table 1 by allowing the additive-control intercept to vary by sector rather than group. The elasticity and measured-price-control intercepts remain sector-specific, and all share-dependent coefficients remain group-specific.

WIOD: 50% Tariff-Price Coverage

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Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	12.68 (4.85)	3.28 (1.13)	7.31 (1.76)	84.3%	3,910.52	591
(2)	Manufacturing	13.86 (5.93)	3.52 (1.19)	7.86 (1.97)	84.9%	2,163.54	535
	Non-manuf. goods	21.25 (15.51)	-0.33 (6.11)	4.78 (3.00)	76.8%	6,465.13	56
(3)	Consumer manuf.	16.45 (6.99)	1.02 (1.53)	5.70 (2.22)	70.8%	1,173.88	260
	Producer manuf.	24.17 (8.20)	4.59 (2.00)	11.81 (2.76)	87.6%	300.94	275
	Non-manuf. goods	21.25 (15.51)	-0.33 (6.11)	4.78 (3.00)	76.8%	6,465.13	56

Notes: This table replicates Table 1 after requiring the named foreign origins entering the shifters to account for at least half of total initial foreign expenditure. Baseline group knot locations and sector classifications are held fixed.

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	13.07 (4.86)	6.42 (3.19)	10.11 (3.80)	86.9%	953.99	563
(2)	Manufacturing	15.35 (6.28)	3.92 (1.49)	8.31 (2.33)	85.9%	1,114.76	523
	Non-manuf. goods	-296.52 (100.53)	91.35 (39.04)	38.62 (40.75)	45.0%	525.65	40
(3)	Consumer manuf.	17.24 (7.60)	1.64 (1.98)	6.08 (2.69)	74.4%	460.89	258
	Producer manuf.	19.59 (7.21)	5.87 (2.20)	12.10 (2.54)	91.7%	156.60	265
	Non-manuf. goods	-296.52 (100.53)	91.35 (39.04)	38.62 (40.75)	45.0%	525.65	40

Notes: This table modifies Table 1 by replacing the endogenous tariff shifter with one constructed from trade-weighted applied tariffs. The agreement instrument retains its baseline definition. Tariff availability reduces the sample, including non-manufacturing from 80 to 40 observations. Baseline knot locations are held fixed.

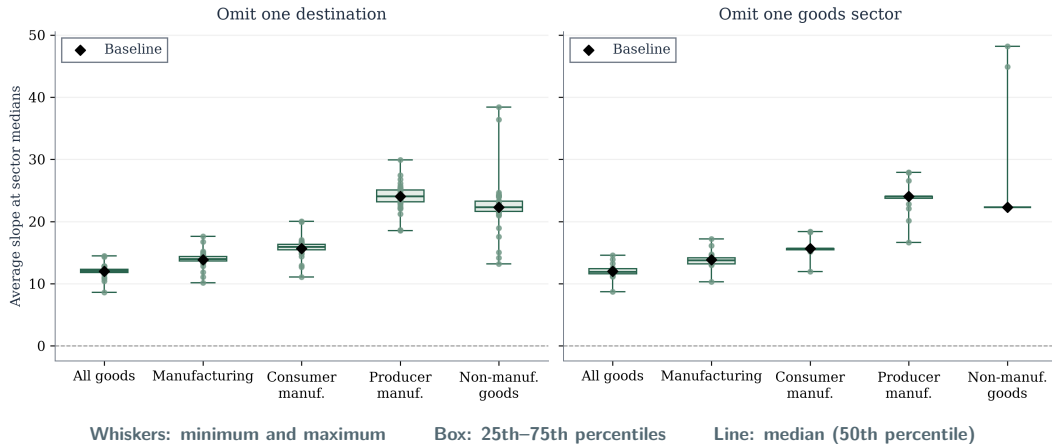
WIOD: Quadratic Elasticity Functions

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Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	13.83 (4.84)	3.15 (1.25)	7.02 (1.81)	82.6%	4,978.40	627
(2)	Manufacturing	16.01 (6.44)	3.49 (1.33)	7.96 (2.23)	85.4%	2,031.77	547
	Non-manuf. goods	22.73 (11.51)	-1.32 (4.63)	5.32 (4.13)	72.5%	221.68	80
(3)	Consumer manuf.	17.31 (7.29)	0.81 (1.92)	5.28 (2.67)	68.4%	1,019.75	269
	Producer manuf.	24.25 (7.45)	4.49 (1.95)	11.68 (2.50)	85.6%	293.80	278
	Non-manuf. goods	22.73 (11.51)	-1.32 (4.63)	5.32 (4.13)	72.5%	221.68	80

Notes: This table modifies Table 1 by replacing the spline with a quadratic in the elasticity schedule, additive controls, and coefficient on measured-price changes. The coefficient structure otherwise remains unchanged.

WIOD: Leave-One-Out Estimates

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Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	4.81	4.89	6.05	100.0%	—	627
(2)	Manufacturing	5.66	5.35	6.76	100.0%	—	547
	Non-manuf. goods	6.17	1.05	2.09	100.0%	—	80
(3)	Consumer manuf.	3.82	3.36	4.35	100.0%	—	269
	Producer manuf.	12.89	7.38	10.32	100.0%	—	278
	Non-manuf. goods	6.17	1.05	2.09	100.0%	—	80

Notes: This table replicates Panel A of Table 1, imposing $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$. Only point estimates are reported; no constrained standard errors or confidence intervals are reported.

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	13.30 (6.04)	4.12 (1.56)	8.35 (2.35)	83.6%	—	627
(2)	Manufacturing	16.60 (9.58)	4.62 (1.75)	9.72 (3.25)	85.4%	—	547
	Non-manuf. goods	24.28 (16.35)	-1.79 (5.66)	5.33 (4.19)	71.2%	—	80
(3)	Consumer manuf.	23.53 (14.42)	0.86 (2.44)	7.30 (4.11)	68.0%	—	269
	Producer manuf.	33.97 (83.19)	25.58 (94.77)	40.78 (135.93)	79.5%	—	278
	Non-manuf. goods	24.28 (16.35)	-1.79 (5.66)	5.33 (4.19)	71.2%	—	80

Notes: This table modifies Panel A of Table 1 by integrating the tariff response along the elasticity schedule, without imposing a positive lower bound. Additive and measured-price controls remain evaluated at the initial share. Parentheses contain destination-clustered standard errors from the nonlinear IV estimator. Conditional SW statistics are not reported for this estimator.

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	4.35	5.00	6.06	100.0%	—	627
(2)	Manufacturing	5.24	5.50	6.81	100.0%	—	547
	Non-manuf. goods	6.46	1.03	2.13	100.0%	—	80
(3)	Consumer manuf.	3.28	3.42	4.29	100.0%	—	269
	Producer manuf.	7.12	7.98	9.67	100.0%	—	278
	Non-manuf. goods	6.46	1.03	2.13	100.0%	—	80

Notes: This table modifies Panel A of Table 1 by integrating the tariff response along the elasticity schedule and imposing $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$. Additive and measured-price controls remain evaluated at the initial share. Only point estimates are reported; conditional SW statistics and constrained standard errors and confidence intervals are not reported.

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	11.09 (3.26)	2.21 (1.17)	6.42 (1.44)	73.2%	2,193.45	627
(2)	Manufacturing	10.86 (4.19)	2.52 (1.38)	6.81 (1.61)	72.6%	1,993.03	547
	Non-manuf. goods	25.26 (18.63)	1.61 (5.47)	9.01 (9.57)	82.5%	53.97	80
(3)	Consumer manuf.	14.08 (5.72)	-0.18 (1.99)	4.92 (1.65)	63.2%	684.42	269
	Producer manuf.	25.24 (8.31)	4.33 (1.68)	11.97 (2.77)	84.9%	114.76	278
	Non-manuf. goods	25.26 (18.63)	1.61 (5.47)	9.01 (9.57)	82.5%	53.97	80

Notes: This table modifies Table 1 by measuring outcomes and observed tariff and price changes over 1996–2007. Regressors and instrument interactions use 1996 domestic shares. Origin weights and knots retain their 1995 definitions, and the agreement instrument retains its 1995–2007 definition.

WIOD: 1996–2007 Lagged-Share Instruments

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Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	11.13 (3.53)	2.42 (1.23)	6.15 (1.46)	73.5%	1,471.83	627
(2)	Manufacturing	11.52 (4.52)	2.69 (1.47)	6.63 (1.67)	74.8%	736.14	547
	Non-manuf. goods	20.83 (19.24)	2.12 (5.73)	7.46 (9.31)	86.2%	35.40	80
(3)	Consumer manuf.	14.47 (5.83)	-0.18 (2.19)	4.32 (1.89)	57.6%	324.28	269
	Producer manuf.	21.80 (7.84)	4.63 (1.71)	11.28 (2.58)	85.3%	171.63	278
	Non-manuf. goods	20.83 (19.24)	2.12 (5.73)	7.46 (9.31)	86.2%	35.40	80

Notes: This table modifies Table 1 by constructing every share-dependent tariff, additive, and measured-price term with the 1996 domestic share and instrumenting it with its 1995-share counterpart. Origin weights and knots are dated 1995, and the agreement instrument retains its 1995–2007 definition. Min. SW F covers all instrumented regressors, including the additive and measured-price share terms.